

1 SUPPLEMENTAL MATERIALS

- 2 Application of Fuzzy Logic in Supplier Selection: A Case Study of ETFE Dome Façade — Tables S1–S11:
- 3 Calculation of Attribute Ratings Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights

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Table S1. Calculation of Attribute 1 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights.

						X				X				X						X			X
0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.01	0.01	0.1	0.00	0.01	0.9	0.01	0.01	0.10	0.10	0.01	0.32	0	0	0	0.316	0.1	0.032	0	0	0	0	0	0
0.04	0.04	0.2	0.01	0.04	0.8	0.03	0.02	0.20	0.20	0.04	0.45	0	0	0	0.447	0.2	0.089	0	0	0	0	0	0
0.09	0.09	0.3	0.027	0.09	0.7	0.063	0.045	0.30	0.30	0.090	0.55	0	0	0	0.548	0.3	0.164	0	0	0	0	0	0
0.16	0.16	0.4	0.064	0.16	0.6	0.096	0.080	0.40	0.40	0.160	0.63	0	0	0	0.632	0.4	0.253	0	0	0	0	0	0
0.25	0.25	0.5	0.125	0.25	0.5	0.125	0.125	0.50	0.50	0.250	0.71	0	0	0	0.707	0.5	0.354	0	0	0	0	0	0
0.36	0.36	0.6	0.216	0.36	0.4	0.144	0.180	0.60	0.60	0.360	0.77	0	0	0	0.775	0.6	0.465	0	0	0	0	0	0
0.49	0.49	0.7	0.343	0.49	0.3	0.147	0.245	0.70	0.70	0.490	0.84	0	0	0	0.837	0.7	0.586	0	0	0	0	0	0
0.64	0.64	0.8	0.512	0.64	0.2	0.128	0.32	0.80	0.80	0.640	0.89	0	0	0	0.894	0.8	0.716	0	0	0	0	0	0
0.81	0.81	0.9	0.729	0.81	0.1	0.081	0.405	0.90	0.90	0.810	0.95	0	0	0	0.949	0.9	0.854	0	0	0	0	0	0
1	1	1	1	1	0	0	0.5	1	1	1	1	0	0	0	1	1	1	0	0	0	0	0	0

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Table S2. Calculation of Attribute 1 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights (continued).

		X			X			X		X		X		X
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.316	0.100	0.032	0.316	0.100	0.032	0.100	0.032	0.100	0.032	0.109	
0.000	0.000	0.000	0.447	0.200	0.089	0.447	0.200	0.089	0.200	0.089	0.200	0.089	0.217	
0.000	0.000	0.000	0.548	0.300	0.164	0.548	0.300	0.164	0.300	0.164	0.300	0.164	0.319	
0.000	0.000	0.000	0.632	0.400	0.253	0.632	0.400	0.253	0.400	0.253	0.400	0.253	0.413	
0.000	0.000	0.000	0.707	0.500	0.354	0.707	0.500	0.354	0.500	0.354	0.500	0.354	0.500	
0.000	0.000	0.000	0.775	0.600	0.465	0.775	0.600	0.465	0.600	0.465	0.600	0.465	0.579	
0.000	0.000	0.000	0.837	0.700	0.586	0.837	0.700	0.586	0.700	0.586	0.700	0.586	0.652	
0.000	0.000	0.000	0.894	0.800	0.716	0.894	0.800	0.716	0.800	0.716	0.800	0.716	0.718	
0.000	0.000	0.000	0.949	0.900	0.854	0.949	0.900	0.854	0.900	0.854	0.900	0.854	0.778	
0.000	0.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.833	

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Table S3. Calculation of Attribute 2 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights.

			X			X		X			X			X	
0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0.01	0.01	0.1	0.001	0.01	0.9	0.009	0.5	0.005	0.01	0.01	0.0001	0	0	0	0.255
0.04	0.04	0.2	0.008	0.04	0.8	0.032	0.5	0.02	0.04	0.04	0.0016	0	0	0	0.27
0.09	0.09	0.3	0.027	0.09	0.7	0.063	0.5	0.045	0.09	0.09	0.0081	0	0	0	0.295
0.16	0.16	0.4	0.064	0.16	0.6	0.096	0.5	0.08	0.16	0.16	0.0256	0	0	0	0.33
0.25	0.25	0.5	0.125	0.25	0.5	0.125	0.5	0.125	0.25	0.25	0.0625	0	0	0	0.375
0.36	0.36	0.6	0.216	0.36	0.4	0.144	0.5	0.18	0.36	0.36	0.1296	0	0	0	0.43
0.49	0.49	0.7	0.343	0.49	0.3	0.147	0.5	0.245	0.49	0.49	0.2401	0	0	0	0.495
0.64	0.64	0.8	0.512	0.64	0.2	0.128	0.5	0.32	0.64	0.64	0.4096	0	0	0	0.57
0.81	0.81	0.9	0.729	0.81	0.1	0.081	0.5	0.405	0.81	0.81	0.6561	0	0	0	0.655

1	1	1	1	1	0	0	0.5	0.5	1	1	1	0	0	0	0.75
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7 **Table S4. Calculation of Attribute 3 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights.**

		X			X			X			X	
0	0	0	0	0	0	0	1	0	0	0	0	0
0.01	0.01	0.0001	0.1	0.01	0.001	0.01	0.9487	0.0095	0.01	0.01	0.0001	0.0822
0.04	0.04	0.0016	0.2	0.04	0.008	0.04	0.8944	0.0358	0.04	0.04	0.0016	0.1468
0.09	0.09	0.0081	0.3	0.09	0.027	0.09	0.8367	0.0753	0.09	0.09	0.0081	0.2079
0.16	0.16	0.0256	0.4	0.16	0.064	0.16	0.7746	0.1239	0.16	0.16	0.0256	0.2717
0.25	0.25	0.0625	0.5	0.25	0.125	0.25	0.7071	0.1768	0.25	0.25	0.0625	0.3414
0.36	0.36	0.1296	0.6	0.36	0.216	0.36	0.6325	0.2277	0.36	0.36	0.1296	0.4184
0.49	0.49	0.2401	0.7	0.49	0.343	0.49	0.5477	0.2684	0.49	0.49	0.2401	0.5030
0.64	0.64	0.4096	0.8	0.64	0.512	0.64	0.4472	0.2862	0.64	0.64	0.4096	0.5946
0.81	0.81	0.6561	0.9	0.81	0.729	0.81	0.3162	0.2561	0.81	0.81	0.6561	0.6899
1	1	1	1	1	1	1	0	0	1	1	1	0.75

8 **Table S5. Calculation of Attribute 4 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights.**

			X			X			X			X			X
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.01	0.01	0.01	0.0001	0	0	0	0.01	0.01	0.0001	0	0	0	0.01	0.01	0.0001
0.04	0.04	0.04	0.0016	0	0	0	0.04	0.04	0.0016	0	0	0	0.04	0.04	0.0016
0.09	0.09	0.09	0.0081	0	0	0	0.09	0.09	0.0081	0	0	0	0.09	0.09	0.0081
0.16	0.16	0.16	0.0256	0	0	0	0.16	0.16	0.0256	0	0	0	0.16	0.16	0.0256
0.25	0.25	0.25	0.0625	0	0	0	0.25	0.25	0.0625	0	0	0	0.25	0.25	0.0625
0.36	0.36	0.36	0.1296	0	0	0	0.36	0.36	0.1296	0	0	0	0.36	0.36	0.1296
0.49	0.49	0.49	0.2401	0	0	0	0.49	0.49	0.2401	0	0	0	0.49	0.49	0.2401
0.64	0.64	0.64	0.4096	0	0	0	0.64	0.64	0.4096	0	0	0	0.64	0.64	0.4096
0.81	0.81	0.81	0.6561	0	0	0	0.81	0.81	0.6561	0	0	0	0.81	0.81	0.6561
1	1	1	1	0	0	0	1	1	1	0	0	0	1	1	1

9 **Table S6. Calculation of Attribute 4 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights (continued).**

				X			X			X	
0	0	0	0	0	0	0	0	0	0	0	0
0.01	0.1	0.1	0.01	0.001	0.01	0.01	0.0001	0.01	0.01	0.0001	0.01
0.04	0.2	0.2	0.04	0.008	0.04	0.04	0.0016	0.04	0.04	0.0016	0.04
0.09	0.3	0.3	0.09	0.027	0.09	0.09	0.0081	0.09	0.09	0.0081	0.09
0.16	0.4	0.4	0.16	0.064	0.16	0.16	0.0256	0.16	0.16	0.0256	0.16
0.25	0.5	0.5	0.25	0.125	0.25	0.25	0.0625	0.25	0.25	0.0625	0.25
0.36	0.6	0.6	0.36	0.216	0.36	0.36	0.1296	0.36	0.36	0.1296	0.36
0.49	0.7	0.7	0.49	0.343	0.49	0.49	0.2401	0.49	0.49	0.2401	0.49
0.64	0.8	0.8	0.64	0.512	0.64	0.64	0.4096	0.64	0.64	0.4096	0.64
0.81	0.9	0.9	0.81	0.729	0.81	0.81	0.6561	0.81	0.81	0.6561	0.81
1	1	1	1	1	1	1	1	1	1	1	1

10 **Table S7. Calculation of Attribute 4 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights (continued).**

[illegible]

11 **Table S8. Calculation of Attribute 5 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights.**

			X			X			X			X	
0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.01	0.01	0.01	0.0001	0.01	0.01	0.0001	0	0	0	0.01	0.01	0.0001	0.0001
0.04	0.04	0.04	0.0036	0.04	0.04	0.0016	0	0	0	0.04	0.04	0.0016	0.0016
0.09	0.09	0.09	0.0081	0.09	0.09	0.0081	0	0	0	0.09	0.09	0.0081	0.0081
0.16	0.16	0.16	0.0256	0.16	0.16	0.0256	0	0	0	0.16	0.16	0.0256	0.0256
0.25	0.25	0.25	0.0625	0.25	0.25	0.0625	0	0	0	0.25	0.25	0.0625	0.0625
0.36	0.36	0.36	0.1296	0.36	0.36	0.1296	0	0	0	0.36	0.36	0.1296	0.1296
0.49	0.49	0.49	0.2401	0.49	0.49	0.2401	0	0	0	0.49	0.49	0.2401	0.2401
0.64	0.64	0.64	0.4096	0.64	0.64	0.4096	0	0	0	0.64	0.64	0.4096	0.4096
0.81	0.81	0.81	0.6561	0.81	0.81	0.6561	0	0	0	0.81	0.81	0.6561	0.6561
1	1	1	1	1	1	1	0	0	0	1	1	1	1

12 Table S9. Calculation of Attribute 5 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights (continued).

[illegible]

13 Table S10. Calculation of Attribute 5 Rating Using Attribute, Sub Attribute, and Sub-Sub Attribute Ratings and Relative Weights (continued).

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